

Strategic Planning and Data Exploration in Logistics

1. Background Research & Project Objectives The logistics sector heavily relies on efficient last-mile delivery to maintain customer satisfaction and control operational costs. Managing a diverse network of third-party delivery partners requires constant monitoring of external factors like weather, vehicle types, and regional infrastructure.

The objective of this project is to analyze a historical logistics dataset (Delivery_Logistics.csv) containing 25,000 delivery records. The primary goal is to uncover the root causes of shipment delays, analyze cost efficiencies across different delivery partners (e.g., DHL, Delhivery, Shadowfax), and predict future delays to optimize supply chain resource allocation.

2. Logistics Scenario Definition This project simulates a **Last-Mile Delivery Optimization Scenario**. As a centralized logistics aggregator, the business dynamically assigns packages to various third-party partners. Currently, the business faces challenges with unpredictable delays caused by adverse weather (rain, fog) and suboptimal vehicle assignments (e.g., using heavy trucks for small electronic packages). By analyzing past performance, the company aims to build a proactive strategy that predicts delays before they happen and routes packages to the most cost-effective and reliable partner.

3. Key Performance Indicators (KPIs) To measure the success and efficiency of the logistics network, this analysis will track the following KPIs:

- **On-Time Delivery (OTD) Rate:** The percentage of shipments successfully delivered without triggering a "delayed" status. This measures overall network reliability.
- **Average Cost Per Kilometer:** The total delivery_cost divided by distance_km. This financial metric will identify which delivery partners and vehicle types offer the best profit margins.
- **Customer Satisfaction Score (CSAT):** The average rating from the delivery_rating column (scale of 1-5). This tracks the direct impact of delivery modes and delays on the end customer.
- **Delay Rate by Weather/Region:** The percentage of late deliveries grouped by specific regional zones and weather conditions to identify operational bottlenecks.

4. Applied Data Science Concepts This project will utilize several core data science methodologies to transform raw data into actionable business strategies:

- **Exploratory Data Analysis (EDA):** Grouping, aggregating, and visualizing data to find hidden trends (e.g., does the ev van vehicle type experience fewer delays than standard trucks in central regions?).
- **Classification (Predictive Modeling):** Machine learning classification algorithms (such as Logistic Regression or Random Forest) will be used to predict the binary outcome of the delayed column (yes/no) based on input features like distance, package weight, and weather.
- **Optimization:** Data-driven logic will be applied to recommend the most optimal delivery partner for a given route to minimize the delivery_cost while maximizing the likelihood of an on-time delivery.

5. End-to-End Strategic Roadmap

The project will be executed over four distinct phases:

- **Phase 1: Strategic Planning (Current):** Define the business problem, outline required KPIs, and set the technical strategy for the analysis.
- **Phase 2: Data Collection & Preprocessing:** Load the Delivery_Logistics.csv dataset using Python. This phase will involve handling missing values, standardizing text data, and formatting the timestamp columns (delivery_time_hours, expected_time_hours) into usable datetime objects for duration calculations.
- **Phase 3: Advanced Analysis & Visualization:** Use libraries like matplotlib and seaborn to build visual dashboards. We will map out correlation matrices to see how distance and package weight impact cost and delivery speed.
- **Phase 4: Predictive Modeling & Recommendations:** Train a machine learning model to forecast delays. The final output will be a set of business recommendations for inventory management and routing optimizations.

6. Code Illustration

Below is the initial Python script designed to load the dataset, perform a basic health check, and calculate the primary KPI (On-Time Delivery Rate):

```
import pandas as pd
import numpy as np

# 1. Data Collection: Load the logistics dataset
file_path = 'Delivery_Logistics.csv'
logistics_data = pd.read_csv(file_path)

# 2. Initial Data Inspection
print("Dataset Overview:")
print(logistics_data.info())

# 3. Calculate KPI: On-Time Delivery Rate
total_shipments = len(logistics_data)
on_time_shipments = len(logistics_data[logistics_data['delayed'] == 'no'])
otd_rate = (on_time_shipments / total_shipments) * 100

print(f"\nStrategic KPI - Overall On-Time Delivery Rate: {otd_rate:.2f}%")

# 4. Calculate KPI: Average Delivery Cost
avg_cost = logistics_data['delivery_cost'].mean()
print(f"Strategic KPI - Average Delivery Cost: ${avg_cost:.2f}")

# 5. Grouping Analysis: Average Rating by Delivery Partner
partner_ratings =
logistics_data.groupby('delivery_partner')['delivery_rating'].mean().reset_index()
```

```
print("\nAverage Customer Rating by Partner:")  
print(partner_ratings.sort_values(by='delivery_rating', ascending=False))
```

7. Conclusion & Expected Outcomes

By implementing this data science roadmap, the logistics organization will transition from a reactive approach to a predictive one. The expected outcomes include a data-backed understanding of which delivery partners perform best under specific conditions, leading to optimized vendor routing. Ultimately, forecasting delays before dispatch will allow the business to proactively alert customers, thereby protecting the brand's reputation, improving the overall CSAT score, and significantly reducing wasted transportation costs.